Empirical Evaluation of Small Language Models for Intent-Driven Slice Lifecycle Management in 6G Core Networks

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Abstract

The adoption of Small Language Models (SLMs) for intent-driven network orchestration in 6G core networks promises low-latency, cost-efficient automation at the edge. However, the trade-offs between model architecture, size, and quantization level remain poorly understood in this domain. This paper presents a systematic empirical evaluation of open-weight SLMs for 6G intent-to-action translation, benchmarking five model families (~8B parameters) and twelve size-quantization configurations of Qwen2.5 (0.5B–7B, Q2_K–Q8_0) on 200 real-world intents across three slice management domains. Our results reveal that Qwen2.5-7B achieves the highest accuracy (29.8% exact match) with the fastest inference (18.7 s/intent) among ~8B models, and that aggressive 2-bit quantization of larger models (7B Q2_K, 29.1%) can outperform higher-bit quantization of smaller models (3B Q8_0, 27.1%) while requiring 2.5x less memory. We identify a clear Pareto frontier and quantify diminishing returns in both size scaling and quantization, providing actionable guidance for deploying SLM-based intent agents in resource-constrained 6G edge environments.

1. Introduction

The transition to 6G networks introduces intent-driven management as a core paradigm, where network operators express desired outcomes in natural language and autonomous agents translate these intents into executable orchestration commands [1–3]. This approach promises reduced operational complexity and faster service provisioning, but places significant demands on the language understanding capabilities of the underlying AI agents.

Large Language Models (LLMs) such as GPT-4 and Llama-70B have demonstrated impressive intent understanding capabilities [4,5]. However, their deployment at the network edge is constrained by latency requirements (sub-100 ms for URLLC), energy budgets, and the absence of GPU accelerators in typical telco infrastructure [6]. Small Language Models (SLMs) — models under 10 billion parameters — offer a compelling alternative, with recent work showing competitive performance on structured understanding tasks when properly prompted [7,8].

Despite growing interest, systematic evaluations of SLMs for 6G intent-driven orchestration remain scarce. Existing studies either focus on a single model family [9], evaluate on synthetic benchmarks unrelated to network operations [10], or neglect the critical impact of post-training quantization on inference performance and accuracy [11].

This paper addresses these gaps through two complementary experiments:

- **Phase 1 — Cross-Family Comparison**: We benchmark five open-weight model families at approximately 8B parameters (Llama 3.1, Qwen 2.5, Gemma 2, GLM-4, DeepSeek-R1) on 200 real-world 6G intents, controlling for quantization (Q4_K_M), hardware, and prompt design.
- **Phase 2 — Size Scaling and Quantization Impact**: Using the best-performing family (Qwen2.5), we evaluate 12 configurations across four model sizes (0.5B, 1.5B, 3B, 7B) and three quantization levels (Q2_K, Q4_K_M, Q8_0), quantifying the accuracy–efficiency trade-off across dimensions.

Our contributions are: (1) the first systematic cross-family SLM benchmark for 6G intent-driven slice management, (2) quantitative evidence for diminishing returns in both model size and quantization precision, and (3) identification of a Pareto-optimal operating point (7B Q2_K) that balances accuracy and resource efficiency for CPU-only edge deployment.

2. Related Work

2.1 Intent-Driven Networking in 6G

Intent-based networking (IBN) has been proposed as a key enabler for zero-touch network and service management (ZSM) in 5G/6G [1]. The ETSI ZSM framework defines intent as "a set of operational goals that a network should meet" [2], while 3GPP has incorporated intent-driven management into Release 18 specifications for network slice lifecycle operations [3]. Recent work has explored LLM-based intent translation for network configuration [4,5], but the computational requirements of large models conflict with the distributed, resource-constrained nature of edge deployments.

2.2 Small Language Models for Domain-Specific Tasks

The emergence of high-quality open-weight SLMs — including Llama 3.1 [12], Qwen 2.5 [13], Gemma 2 [14], and GLM-4 [15] — has enabled on-device and edge-deployed language understanding. These models, ranging from 0.5B to 9B parameters, achieve competitive performance on structured extraction tasks when fine-tuned or carefully prompted [7,8]. However, their application to network intent understanding remains underexplored.

2.3 Quantization and Model Compression

Post-training quantization (PTQ) reduces model size and inference latency by representing weights with lower bit precision [16]. The K-quant family (Q2_K–Q8_0) provides a spectrum of compression ratios, with Q4_K_M serving as a common default [17]. While the accuracy degradation from quantization has been studied on standard NLP benchmarks [10,11], domain-specific tasks involving structured output (e.g., JSON generation for network commands) may exhibit different sensitivity profiles. Our work is the first to characterize quantization impact on structured intent-to-action translation in 6G.

3. Methodology

3.1 Dataset

We construct a dataset of 200 intents spanning three 6G slice management domains:

- **Slicing Provisioning** (n=70): Creation of new network slices with specified QoS parameters (e.g., "Create a URLLC slice with 1ms latency for autonomous vehicle platooning in Bangkok").
- **Scaling Request** (n=70): Modification of existing slice capacity (e.g., "Scale the eMBB slice for stadium event to 10 Gbps during the concert").
- **Conflict Resolution** (n=60): Resolution of resource conflicts between competing slice requests (e.g., "URLLC factory slice needs priority over eMBB streaming slice during emergency").

Each intent is annotated with a ground-truth JSON action containing `action`, `slice_type`, and domain-specific parameters. Intents span three complexity levels (Simple, Medium, Complex) to test robustness.

3.2 Models

Phase 1 evaluates five model families at approximately 8B parameters, all using Q4_K_M quantization for fair comparison:

Model	Family	Parameters	Quantization
Llama 3.1 8B	Meta	8.0B	Q4_K_M
Qwen 2.5 7B	Alibaba	7.6B	Q4_K_M
Gemma 2 9B	Google	9.4B	Q4_K_M
GLM-4 9B	Tsinghua	9.4B	Q4_K_M
DeepSeek-R1 8B	DeepSeek	8.0B	Q4_K_M

Note: Gemma 2 and GLM-4 default Ollama tags use Q4_0 quantization; we created custom models from HuggingFace GGUF Q4_K_M files to ensure fairness.

Phase 2 evaluates Qwen2.5 across four sizes and three quantization levels, yielding 12 configurations:

Size	Parameters	Q2_K size	Q4_K_M size	Q8_0 size
0.5B	494M	338 MB	397 MB	531 MB
1.5B	1.5B	~650 MB	986 MB	~1.6 GB
3B	3.1B	1.3 GB	1.9 GB	3.1 GB
7B	7.6B	~4 GB	4.7 GB	~8 GB

3.3 Evaluation Protocol

All experiments run on identical hardware: Proxmox VMs with 16 CPU cores and 31 GB RAM, using Ollama as the inference engine. The system prompt is uniform across all runs, instructing models to output valid JSON only. Each intent is evaluated once (no sampling) with temperature=0.1 and `num_predict=1024`.

Metrics: - **Format Accuracy** (%): Proportion of responses that are valid JSON with an `action` field. - **Action Correct** (%): Proportion where the `action` value matches ground truth. - **Exact Match** (%): Average Jaccard similarity between parsed JSON keys/values and ground truth. - **Inference Time** (s/intent): Wall-clock inference duration. - **Throughput** (tokens/s): Tokens generated per second.

3.4 Prompt Design

The system prompt instructs the model to act as a "6G Core Network Slice Management Agent," with explicit rules: JSON-only output, no markdown, and appropriate use of slice types (eMBB, URLLC, mMTC). A one-shot example is provided. This enriched prompt design was validated in prior work [9] and shown to improve format compliance.

4. Phase 1: Cross-Family Comparison

4.1 Overall Performance

Table 1 presents the aggregate results for all five ~8B models. DeepSeek-R1 is excluded from analysis due to a known incompatibility: its chain-of-thought reasoning mechanism generates unbounded `thinking` tokens on CPU-only inference, resulting in infinite generation loops and 0% format validity. This finding highlights a critical limitation of reasoning-oriented architectures for latency-bounded edge deployment.

Table 1: Phase 1 — Cross-Family Performance at ~8B (Q4_K_M)

Model	Format	Action	Exact	Time (s)	Tok/s
Qwen 2.5 7B	100.0%	43.5%	29.8%	18.7	2.2
Gemma 2 9B	98.5%	45.5%	29.3%	39.8	1.0
Llama 3.1 8B	100.0%	36.5%	26.0%	19.1	2.2
GLM-4 9B	100.0%	29.0%	24.6%	22.9	1.8
DeepSeek-R1 8B	0.0%	0.0%	0.0%	77.9	3.3

Qwen2.5-7B achieves the highest exact match (29.8%) while maintaining the fastest inference (18.7 s/intent). Gemma2-9B closely follows in accuracy (29.3%) but at 2.1x the inference cost (39.8 s/intent). Llama3.1-8B provides competitive speed (19.1 s/intent) but trails in exact match by 3.8 percentage points. GLM-4-9B underperforms across all accuracy dimensions despite comparable model size, suggesting suboptimal alignment with the JSON-structured output format.

Format accuracy is near-perfect (>=98.5%) for all non-reasoning models, indicating that the enriched prompt design effectively constrains output structure.

4.2 Domain-Level Analysis

Table 2: Phase 1 — Action Correct by Domain (%)

Model	Provisioning	Scaling	Conflict
Qwen 2.5 7B	100.0%	18.6%	6.7%
Gemma 2 9B	97.1%	25.7%	8.3%
Llama 3.1 8B	87.1%	11.4%	6.7%
GLM-4 9B	54.3%	21.4%	8.3%

All models excel at Slicing Provisioning (Qwen: 100%), where the task is a straightforward CREATE action. Performance drops sharply for Scaling Request (11.4–25.7%) and Conflict Resolution (6.7–8.3%), which require nuanced understanding of resource constraints and priority semantics. This domain difficulty gradient is consistent across model families, suggesting it is inherent to the task complexity rather than model-specific weaknesses.

4.3 Key Findings

1. **Qwen2.5-7B dominates the Pareto frontier**: highest accuracy at fastest speed among all ~8B models.
2. **Gemma2-9B trades speed for marginal accuracy gain in action prediction** (45.5% vs 43.5%), making it suitable only when inference latency is not critical.

3. **DeepSeek-R1 is incompatible with CPU-only edge inference**: its chain-of-thought mechanism produces unbounded token generation, a critical limitation for latency-bounded 6G applications.
4. **All models struggle with complex intents**: Conflict Resolution accuracy remains below 10% across families, indicating that current SLMs lack the reasoning depth required for multi-constraint resource arbitration.

5. Phase 2: Size Scaling and Quantization Impact

5.1 Experimental Design

Based on Phase 1 results, we select Qwen2.5 as the representative architecture and evaluate 12 configurations: 4 model sizes (0.5B, 1.5B, 3B, 7B) x 3 quantization levels (Q2_K: 2-bit, Q4_K_M: 4-bit, Q8_0: 8-bit). This full-factorial design enables isolation of the independent effects of parameter count and weight precision on intent understanding performance.

5.2 Accuracy vs. Model Size

Table 3: Phase 2 — Exact Match (%) by Size and Quantization

Size	Q2_K	Q4_K_M	Q8_0	Delta (Q2-Q8)
0.5B	18.6%	18.7%	19.7%	+1.1 pp
1.5B	21.0%	26.3%	26.5%	+5.5 pp
3B	24.0%	27.5%	27.1%	+3.1 pp
7B	29.1%	29.8%	30.4%	+1.3 pp

Three key patterns emerge:

1. Diminishing returns in size: The marginal gain per parameter doubling decreases sharply. Moving from 0.5B to 1.5B (3x parameters) yields +7.6 pp exact match, while 1.5B - 3B (2x) yields only +1.2 pp, and 3B - 7B (2.3x) yields +2.3 pp. The 3B model achieves 92% of 7B accuracy at 2.5x smaller size.

2. Non-monotonic quantization sensitivity: The impact of quantization varies with model size. At 0.5B and 7B, Q2_K degrades accuracy by only 1.1 pp compared to Q8_0, suggesting these sizes are robust to aggressive compression. At 1.5B, the gap widens to 5.5 pp, indicating higher sensitivity. At 3B, intermediate sensitivity (3.1 pp) is observed.

3. Cross-size Pareto dominance: The 7B Q2_K configuration (29.1% exact match, 12.4 s/intent) outperforms 3B Q8_0 (27.1%, 12.9 s/intent) in both accuracy and speed while occupying approximately 4.0 GB vs 3.1 GB on disk. This challenges the conventional wisdom that higher-bit quantization of smaller models is always preferable to lower-bit quantization of larger models.

5.3 Inference Efficiency

Table 4: Phase 2 — Inference Time and Throughput

Size	Q2_K time	Q4_K_M time	Q8_0 time	Q2_K tok/s	Q8_0 tok/s
0.5B	4.3s	3.8s	3.8s	9.1	8.5
1.5B	3.4s	8.5s	8.7s	9.3	5.4
3B	8.3s	11.9s	12.9s	4.4	2.9
7B	12.4s	18.7s	16.0s	3.3	2.5

Inference time generally increases with model size, but quantization introduces interesting non-linearities. At 7B, Q8_0 (16.0 s/intent) is actually faster than Q4_K_M (18.7 s/intent), likely because 8-bit weights eliminate dequantization overhead during CPU inference, offsetting the larger model footprint. The 1.5B model shows a pronounced quantization cliff: Q4_K_M and Q8_0 are 2.5x slower than Q2_K (8.5–8.7s vs 3.4s), consistent with the model crossing a cache-size threshold where the larger quantized model no longer fits in L3 cache.

Throughput (tokens/s) declines with both size and quantization precision, from 10.3 tok/s (0.5B Q4_K_M) to 2.2 tok/s (7B Q4_K_M). The 1.5B Q2_K configuration achieves the highest throughput (9.3 tok/s) among models with >20% exact match, representing a favorable operating point for throughput-sensitive deployments.

5.4 Format Accuracy Trends

Format accuracy (valid JSON output) is generally high across all configurations (>=95%), with a slight upward trend at larger sizes. The 0.5B models show the most variability (95.0–100.0%), while all 7B configurations achieve perfect format compliance (100.0%). This suggests that even the smallest tested models can learn structured output formatting when given a sufficiently explicit prompt, though consistency improves with scale.

5.5 Pareto Analysis

Figure 1 (see `phase2_pareto_exact_vs_time.png`) plots all 12 configurations in the accuracy–speed space, revealing a clear Pareto frontier. The frontier is defined by: 0.5B Q8_0 (19.7%, 3.8s) - 1.5B Q4_K_M (26.3%, 8.5s) - 7B Q2_K (29.1%, 12.4s) - 7B Q8_0 (30.4%, 16.0s). The 1.5B Q2_K point (21.0%, 3.4s) is also notable for providing the fastest inference among models with >20% accuracy.

6. Discussion

6.1 Practical Implications for 6G Edge Deployment

Our results provide several actionable guidelines for deploying SLM-based intent agents in 6G core networks:

1. **Qwen2.5-7B with Q4_K_M** represents the best overall balance of accuracy and speed for general deployment, achieving 29.8% exact match at 18.7 s/intent.
2. **For memory-constrained edge nodes**, Qwen2.5-7B with Q2_K quantization (29.1%, 4.0 GB) provides near-identical accuracy with 2.5x less memory than the Q8_0 variant, at the cost of only 1.3 pp exact match.
3. **For ultra-low-latency applications**, Qwen2.5-1.5B with Q2_K (21.0%, 3.4 s/intent) achieves the fastest inference among usable models, though with a significant accuracy penalty compared to larger variants.
4. **Qwen2.5-3B at Q4_K_M** (27.5%, 1.9 GB) is the sweet spot for resource-constrained deployments, delivering 92% of 7B accuracy at 40% of the memory footprint.

6.2 Limitations

Several limitations should be acknowledged:

- **CPU-only inference**: All experiments run on CPU (16-core VM). GPU-accelerated inference would alter the speed–accuracy trade-off, particularly for larger models, though many 6G edge nodes lack GPU hardware.
- **Resource metrics**: This study does not report real-time CPU utilization, memory bandwidth, or energy consumption during inference. These metrics are critical for edge deployment planning and will be addressed in future work with integrated system monitoring.
- **Single-prompt evaluation**: Results reflect a single prompt design. Prompt engineering, few-shot examples, and fine-tuning may yield different relative rankings.
- **Domain specificity**: The 200-intent dataset, while covering three real-world domains, may not generalize to all 6G intent types. Larger and more diverse benchmarks are needed.

6.3 Comparison with Prior Work

Our findings align with the broader SLM literature showing that Qwen2.5 models punch above their weight class on structured tasks [13], and that K-quant quantization produces graceful degradation curves [17]. The cross-size Pareto dominance we observe (7B Q2_K > 3B Q8_0) has not been previously reported in the network orchestration domain and warrants further investigation across different model families and tasks.

7. Conclusion

This paper presented a systematic empirical evaluation of Small Language Models for intent-driven slice lifecycle management in 6G core networks. Across two controlled experiments — cross-family comparison at ~8B and size-quantization scaling of Qwen2.5 — we demonstrated that:

1. Qwen2.5-7B with Q4_K_M quantization achieves the best accuracy–speed trade-off among five open-weight model families (29.8% exact match, 18.7 s/intent).
2. Aggressive 2-bit quantization (Q2_K) of larger models can outperform higher-bit quantization of smaller models (7B Q2_K: 29.1% vs 3B Q8_0: 27.1%), challenging conventional deployment heuristics.
3. Diminishing returns are evident in both model size (3B captures 92% of 7B accuracy) and quantization precision (Q4-Q8 gains <2 pp at all sizes).
4. Reasoning-oriented architectures (DeepSeek-R1) are currently incompatible with CPU-only edge inference due to unbounded chain-of-thought generation.

These findings provide a quantitative foundation for selecting and configuring SLMs for 6G intent-driven network management. Future work should extend the evaluation to GPU-accelerated inference, integrate energy and resource monitoring, and explore fine-tuning strategies tailored to network orchestration domains.

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All experimental data, code, and figures available at the project repository.